FLEET TEAM DIGEST

737MAX-FTD-32-17008

Issue Title : Prevention of Wheel and Brake Fires

Airplane Model	ATA	Minor Model(s)
737MAX	3246-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/02/2017	05/02/2017	04/19/2017

Estimated Completion Date	Next Update Date

Applicability

All 737MAX airplanes.

Description

Boeing received a report of a fire at one of the main gear wheel/brake installations on a 737-900ER airplane equipped with carbon brakes. The fire occurred one day after a wheel/tire replacement.

The cause of the fire was attributed to excessive grease in the brake's torque tube cavity between the inboard side of the wheel and brake. Since this brake does not have a lubrication fitting at the piston housing, we suspect that the excess grease came from a liberally packed wheel bearing or an over-application of grease on the axle sleeve and/or brake sleeve during the wheel installation.

Background

Typically, the cause of wheel and brake fires can be attributed one of the following:

- The ignition of excessive grease that has accumulated in the brake assembly cavity and/or on the axle.

- The ignition of residual cleaning fluids can be retained by some wheel heat shields after being saturated with flammable solvents during maintenance.

- The ignition of hydraulic fluid that is on or near hot parts of the brake that is associated with a leaks or spill.

A fire will most likely occur within the next few landing cycles when the brake assembly reaches a temperature that ignites the fuel source.

Airplanes with carbon brakes are more susceptible to these types of fires due to the higher operating temperatures associated with carbon brakes.

The average flash point of the recommended wheel bearing greases (Aeroshell 22, Mobil Grease 28, and Mobil Aviation Grease SHC 100) used to lubricate the piston housing bushing is 435 degrees F.

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Status

Closed

Interim Action

None

Final Action

Boeing has published the referenced Maintenance Tip to advise operators about the possibility of fires in the wheel and brake areas, their causes, and how to prevent them. In addition, Boeing discussed this subject in the 2nd quarter 2007 edition of Boeing Aero Magazine.

The wheel and brake installation sections of the AMM provide cautions about applying excessive grease to the axle sleeve and brake sleeve axle due to the risk of fire.

Boeing has developed a training aid to help maintenance personnel visualize and understand proper wheel/axle greasing and cleaning techniques. This aid is a DVD (digital video disc) titled, “Main Landing Gear Wheel/Brake Area Fire Prevention: Maintenance TIPS”. Boeing recommends operators consider showing this DVD to Engineering and Maintenance personnel associated with landing gear duties during crew meetings. This DVD is available from Boeing Data and Services Management. The DVD designator is: VPS48559.

Operator Action

It is important to remove the old grease from the axle first. Use of a dry rag is recommended since cleaning fluids and solvents can cause damage to carbon brakes and titanium components.

While it is important to have adequate lubrication within the wheel bearings, care must be taken when lubricating the axle and brake sleeves to avoid applying excess amounts of grease since the excess grease can accumulate in the hot areas of the wheel and brake installation. Only a thin layer of grease is necessary at the wheel/axle interface for wheel/tire installations. Similarly, only a thin layer of grease is necessary at the interface surfaces of the brake and axle sleeves during brake installation.

When applying grease to the axle bushings on the brake assembly, it is important to completely fill the grooves in the bushings with grease.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Maintenance Tip (MT)	737 MT 32-010	1	
Aircraft Maintenance Manual (AMM)	32-41-41/401		
Aircraft Maintenance Manual (AMM)	32-45-11/401		

Related Categories

- In Service Occurrences : null

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